

CLASS XI · LECTURE 6

Data Communication

Two periods · Signals, modes, modulation, media, synchronous transmission

By the end of this lecture you can

- Name the five components of a data-communication system.
- Distinguish analog and digital signals, and modulation and demodulation.
- Define simplex, half duplex and full duplex with an example each.
- Compare guided and unguided media, and synchronous and asynchronous transmission.

A message has to travel**DEFINITION Data communication**

Data communication is the exchange of data between two devices through some medium. The exchange follows a set of rules called a protocol.

Five components come up whenever the question says “components of data communication”. Miss one and the answer looks unfinished. They are sender, receiver, medium, message and protocol.

Component	Role	Example
Sender	The device that starts the message	Your computer
Receiver	The device that is meant to get it	A friend's computer, or a server
Medium	The path the signal uses	A cable, or radio
Message	The data being sent	A file, a mail, a voice
Protocol	The agreed rules of format, speed and order	TCP/IP, HTTP, SMTP

Analog and digital signals

An **analog** signal varies continuously, like a smooth wave. A spoken voice on an old telephone line is analog. A **digital** signal jumps between separate levels, usually written as 0 and 1. Data inside a computer is digital. A keyboard press becomes a digital code. Converting between the two is the modem's whole job, and it is a separate pair of words from “synchronous”.

ON THE BOARD Modulation and demodulation

Modulation puts a digital signal onto an analog carrier so it can travel on a line made for analog waves. Demodulation takes the analog signal and recovers the digital data. A modem modulates when it sends and demodulates when it receives. The name is modulator–demodulator. The one-mark statement “converting an analog signal to a digital signal” is demodulation.

Direction of data: three modes

Mode	Direction	Everyday example
Simplex	One direction only. The other end cannot reply on that channel	A keyboard sending to the CPU, a television broadcast, a loudspeaker
Half duplex	Both directions, but only one direction at a time	A walkie-talkie: you talk, then you listen
Full duplex	Both directions at the same time	A telephone call

EXAM TIP The half-duplex wording

The paper's phrase is almost fixed: a mode that allows information to travel in two directions, but not at the same time, is half duplex. If the question says "at the same time", the answer is full duplex. If it says "one direction only", the answer is simplex.

Synchronous and asynchronous transmission

This contrast is about **timing**, not about analog and digital. Do not mix the two tables in one answer.

Point	Synchronous	Asynchronous
Clock	Sender and receiver share a timing signal	No shared clock
How bytes are framed	Data moves as a block	Each character has a start bit and a stop bit
Speed	Faster, less extra framing	Slower, because of start and stop bits
Where you meet it	High-speed links	Keyboards and older serial ports

Channels and media

A Section C question, "explain data communication channels with examples", is answered as transmission media. Split the page into guided and unguided. Guided means a physical wire or fibre. Unguided means a wave through air or space.

Medium	Guided?	One fact worth a mark
Twisted pair	Yes	Two copper wires twisted to reduce noise. Cheap. Used in telephone and LAN cables.
Coaxial cable	Yes	A copper core, insulation, and a metal shield. Better against noise than twisted pair. Used in cable TV and older LANs.
Fibre optic	Yes	Light in a glass thread. Very fast, long distance, no electrical noise. More costly to join.
Radio waves	No	Broadcast through the air. Need no cable. Can be interfered with.
Microwave	No	Straight-line towers, a few tens of kilometres apart. Used for long links.
Satellite	No	A microwave repeater in orbit. Reaches remote areas. There is a delay.
Infrared	No	Short range, line of sight. A remote control.

Point-to-point communication joins one sender to one receiver, like a private phone call or a cable between two computers. **Broadcast** communication sends one message to many receivers at once, like radio, or a hub flooding a frame onto a bus. The paper asks for this difference in Section B. Two lines and one example on each side are enough.

Signals weaken and get dirty on the way. **Attenuation** is the loss of strength with distance, cured by a repeater or amplifier. **Noise** is an unwanted signal added on the way. You can mention both if a question says "impairments", but they are not required in every media answer.

Checkpoint

Try the questions before you read the model answers. Section B answers on the real paper should stay within about six or seven lines.

Multiple choice

1. Converting an analog signal into a digital signal is:

- A) Modulation
- B) Demodulation
- C) Compilation

- D) Encryption
2. Two directions, but not at the same time, describes:
- A) Simplex
 - B) Half duplex
 - C) Full duplex
 - D) Broadcast only
3. Coaxial cable and fibre optic are examples of:
- A) A router
 - B) Communication media
 - C) A modem
 - D) System software

Short answers

- S1.** Name the components of data communication.
- S2.** Differentiate analog signals and digital signals.
- S3.** Differentiate synchronous and asynchronous transmission.

Detailed answer

- L1.** Explain data communication channels with examples.

Model answers

Cover this section while you attempt the checkpoint. The wording is a full-mark style answer, not the only acceptable wording.

MCQ 1. B. Modulation is the opposite direction: digital onto an analog carrier.

MCQ 2. B. A walkie-talkie. A telephone call is full duplex.

MCQ 3. B. They are paths for the signal, not devices that choose a route.

S1. The five components are the sender, which starts the message; the receiver, which is meant to get it; the medium, which is the path; the message, which is the data; and the protocol, which is the set of rules both sides follow. A protocol example is HTTP or SMTP.

S2. An analog signal varies continuously, like a voice wave on a telephone line. A digital signal has separate levels, usually 0 and 1, like data inside a computer. A modem converts between them: modulation sends digital data as an analog signal, and demodulation recovers the digital data.

S3. In synchronous transmission the sender and receiver share a clock and data moves as a block, which is fast. In asynchronous transmission there is no shared clock; each character is framed by a start bit and a stop bit, which is slower. Keyboards use the asynchronous method.

L1. A communication channel is the medium that carries the signal. Guided media use a physical path. Twisted-pair cable is two copper wires twisted together; it is cheap and common in LANs and telephone lines. Coaxial cable has a shielded copper core and resists noise better; cable television uses it. Fibre-optic cable carries light in glass and gives high speed over long distances. Unguided media use waves. Radio waves broadcast through the air. Microwaves travel in a straight line between towers. A satellite repeats a microwave signal and can reach remote places, with a small delay. Infrared is short-range and needs line of sight, as in a remote control. The choice depends on distance, speed, cost and whether a cable can be laid.